7.4 Solving Polynomial Equations in Factored Form

Name _____ Period ____ Date _____

Algebra I 7.4 · Student sheet

I can…

- use the Zero-Product Property to solve equations.
- factor out the greatest common monomial factor of a polynomial.
- solve polynomial equations by factoring out the GCF.

Vocabulary · fill in the missing word

Factored form — a _____ written as a product of factors, like $(x - 4)(x + 1)$

Zero-Product Property — if $ab =$ _____, then $a = 0$ or $b = 0$

Roots — the solutions of a _____ equation

Repeated roots — a root that comes from a factor that _____ more than once

Part 1 · Zero hunt

Two numbers with a product of 12	Two numbers with a product of 0

When a product is 0, what must be true? _____

Graph $y = (x - 2)(x + 3)$ in Desmos. Where does it cross the x-axis? _____

How do those x-values relate to the factors?

Part 2 · Notes: fill in the blanks

Zero-Product Property: if $ab = 0$, then _____ or _____.

To solve a factored equation, set _____ factor equal to 0.

A root that comes from a factor that appears more than once is a _____ root.

To solve by factoring: get _____ on one side, _____, then use the Zero-Product Property.

Don’t divide both sides by x: you lose the solution _____.

Part 3 · Examples (work them with the class)

Example 1

Solve the equation.

$(x - 4)(x + 1) = 0$

Example 2

Solve the equation.

$(3x + 2)(x - 5) = 0$

Example 3

Solve the equation.

$(n - 2)^2(n + 7) = 0$

Example 4

Factor the polynomial.

$12x^3 - 18x^2$

Example 5

Solve the equation.

$4x^2 = 20x$

Example 6

The arch of a bridge can be modeled by $y = -0.2(x - 4)(x - 24)$, where x and y are measured in feet and the x-axis is the water.

How wide is the arch at the water?

Part 4 · Partner practice

Partner 1

Solve the equation.

$(2x - 7)(x + 3) = 0$

Partner 2

Solve the equation.

$9x^2 = 15x$